

1200. Minimum Absolute Difference

Solved 

Easy

 Topics

 Companies

 Hint

Given an array of **distinct** integers `arr`, find all pairs of elements with the minimum absolute difference of any two elements.

Return a list of pairs in ascending order(with respect to pairs), each pair `[a, b]` follows

- `a, b` are from `arr`
- `a < b`
- `b - a` equals to the minimum absolute difference of any two elements in `arr`

Example 1:

Input: `arr = [4,2,1,3]`

Output: `[[1,2],[2,3],[3,4]]`

Explanation: The minimum absolute difference is 1. List all pairs with difference equal to 1 in ascending order.

Example 2:

Input: `arr = [1,3,6,10,15]`

Output: `[[1,3]]`

Example 3:

Input: `arr = [3,8,-10,23,19,-4,-14,27]`

Output: `[[ -14, -10], [19, 23], [23, 27]]`

Constraints:

- `2 <= arr.length <= 105`
- `-106 <= arr[i] <= 106`

Python:

```
class Solution:
    def minimumAbsDifference(self, A: List[int]) -> List[List[int]]:
        A.sort()
        n = len(A)
        minDiff = 2e6 + 1
        res = []

        for i in range(1, n):
            diff = A[i] - A[i - 1]
            if diff < minDiff:
                minDiff = diff
                res = [[A[i - 1], A[i]]]
            elif diff == minDiff:
                res.append([A[i - 1], A[i]])

        return res
```

Javascript:

```
const minimumAbsDifference = arr => {
    arr.sort((a, b) => a - b);
    let minDiff = Infinity;
    let temp = [];

    for (let i = 1; i < arr.length; i++) {
        const diff = arr[i] - arr[i - 1];
        if (diff < minDiff) {
            minDiff = diff;
            temp = [i - 1];
        } else if (diff === minDiff)
            temp.push(i - 1);
    }

    return temp.map(i => [arr[i], arr[i + 1]]);
};
```

Java:

```
class Solution {
    public List<List<Integer>> minimumAbsDifference(int[] A) {
        Arrays.sort(A);
        int n = A.length;
        int minDiff = Integer.MAX_VALUE;
        List<List<Integer>> res = new ArrayList<>();
```

```
for (int i = 1; i < n; i++) {  
    int diff = A[i] - A[i - 1];  
    if (diff < minDiff) {  
        minDiff = diff;  
        res = new ArrayList<>();  
        res.add(Arrays.asList(A[i - 1], A[i]));  
    } else if (diff == minDiff) {  
        res.add(Arrays.asList(A[i - 1], A[i]));  
    }  
}  
  
return res;  
}
```